

Learning-Guided Simulated Annealing for the Capacitated Vehicle Routing Problem

Anonymous submission

Abstract

Introduction

Background and Related Work

Problem Definition

The *Capacitated Vehicle Routing Problem (CVRP)* is defined on a complete undirected graph $G = (V, E)$, where $V = \{0\} \cup \mathcal{C}$ and $E = V \times V$ denote the sets of nodes and edges, respectively. Node 0 corresponds to the central depot, and $\mathcal{C} = \{1, \dots, N\}$ represents the set of N customers.

Each edge $(i, j) \in E$ is associated with a travel cost $c_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$, the Euclidean distance between the 2D coordinates $\mathbf{p}_i, \mathbf{p}_j \in [0, 1]^2$, satisfying $c_{ij} = c_{ji}$. Each customer $i \in \mathcal{C}$ has a non-negative demand q_i , while the depot has zero demand ($q_0 = 0$). An unlimited fleet of identical vehicles, each with uniform capacity Q , is stationed at the depot; the number of routes m is therefore a decision variable unconstrained from above.

A route $R_k = (0, v_1, v_2, \dots, v_p, 0)$ is a simple cycle starting and ending at the depot, where $\{v_1, \dots, v_p\} \subseteq \mathcal{C}$ denotes the subset of customers served by vehicle k . A *solution* $s = \{R_1, \dots, R_m\}$ consists of a set of m routes. A solution s is *feasible* if and only if it satisfies the following conditions:

1. Each customer is visited exactly once by a single vehicle,

$$\sum_{k=1}^m \mathbf{1}_{\{i \in R_k\}} = 1, \quad \forall i \in \mathcal{C}.$$

2. For every route R_k , the total demand of its customers does not exceed the vehicle capacity Q , i.e.,

$$\sum_{i \in R_k \setminus \{0\}} q_i \leq Q, \quad \forall k = 1, \dots, m.$$

The set of *feasible solutions* is denoted by $\mathcal{F}$.

Let $x_{ij}(s)$ be a binary indicator variable with $x_{ij}(s) = 1$ if edge (i, j) is traversed in any route R_k of s , and $x_{ij}(s) = 0$

otherwise. The objective of the CVRP is to find a feasible solution $s^* \in \mathcal{F}$ that minimizes the total travel cost $C(s)$:

$$s^* = \arg \min_{s \in \mathcal{F}} \left\{ C(s) = \sum_{i \in V} \sum_{j \in V, i < j} c_{ij} x_{ij}(s) \right\}.$$

Proposed Method

Overview

We propose *Learning-Guided Simulated Annealing (LG-SA)*, a hybrid method that combines *Simulated Annealing (SA)* with a lightweight neural policy trained by *Reinforcement Learning (RL)*. Following the Neural Simulated Annealing framework (Correia, Worrall, and Bondesan 2023), the SA component orchestrates the global exploration of the solution space through its acceptance criterion and cooling schedule, while the neural network replaces the single blind component of SA — the uniform move proposal — with a learned, state-dependent proposal distribution. This design contrasts with “heavyweight” improvement architectures such as the Dual-Aspect Collaborative Transformer (Ma et al. 2021), which replace the heuristic logic entirely with a deep encoder: our policy remains deliberately small, so that the guided search retains a per-step cost close to that of classical SA and can be batched over thousands of instances on a single GPU.

Simulated Annealing (Kirkpatrick, Gelatt, and Vecchi 1983) iteratively explores the neighborhood of a current solution. Given a solution s , SA samples a neighboring solution $s' \in \mathcal{N}(s)$ and accepts it according to the *Metropolis criterion*: improving moves are always accepted, while worsening moves are accepted with probability $\exp(-\Delta C/T)$, where $\Delta C = C(s') - C(s)$ is the cost difference and T the current temperature. The temperature decays from an initial value T_0 to a final value T_{final} over the search horizon according to a monotone cooling schedule, which regulates the transition from exploration to exploitation.

Neighboring solutions are produced by a *neighborhood operator* $H : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ that maps a solution s and an action $a \in \mathcal{A}(s)$ to the neighbor $s' = H(s, a)$, so that

$$\mathcal{N}(s) = \{H(s, a) : a \in \mathcal{A}(s)\}.$$

In our setting, an action designates a pair of nodes $a = (i, j)$, and H applies a standard pairwise local search move to them (e.g., relocating node i next to node j). The operator is a

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Algorithm 1 Learning-Guided Simulated Annealing

Input: instance, initial solution s , policy π_θ , operator H **Output:** best solution found s^*

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1: Initialize  $s^* \leftarrow s$  and temperature  $T \leftarrow T_0$ 
2: while termination criterion not met do
3:   Encode  $s$  into per-node feature vectors
4:   Sample action  $a \sim \pi_\theta(\cdot | s)$  (guided proposal)
5:   Generate candidate  $s' \leftarrow H(s, a)$ 
6:    $s \leftarrow s'$  with probability  $\min(1, e^{-\Delta C/T})$  (Metropolis)
7:   Update  $s^*$  if  $C(s) < C(s^*)$ ; cool  $T$ 
8: end while
9: return  $s^*$ 
```

modular component of the framework; the candidate operators and their comparison are presented in the experimental section.

Whereas classical SA draws a uniformly over $\mathcal{A}(s)$, LG-SA samples it from a policy π_θ conditioned on the current solution:

$$a \sim \pi_\theta(\cdot | s) \quad \text{and} \quad s' = H(s, a) \in \mathcal{N}(s).$$

Everything else — the operator, the Metropolis acceptance, the cooling schedule — is kept from classical SA. The overall procedure is summarized in Algorithm 1 and illustrated in Figure 1.

The Guided Search as a Markov Decision Process

To learn the proposal distribution with RL, we frame the SA loop as a Markov Decision Process. At step t , the *state* s_t is the current feasible solution together with the search meta-information (current temperature and remaining budget), encoded as a set of per-node feature vectors described below. The *action* $a_t = (i, j)$ is the node pair handed to the operator H . The *transition* is the composition of the operator and the stochastic Metropolis acceptance: the next state is $H(s_t, a_t)$ if the move is accepted and s_t otherwise, so the environment dynamics include the acceptance randomness of SA.

The *reward* is derived from the evolution of the solution cost along the search: a move that is accepted and improves the solution receives a positive reward proportional to the (normalized) cost reduction, a rejected move yields no improvement signal, and proposals leading to infeasible neighbors can be penalized. Reward variants that instead emphasize improvements of the best solution found so far, or the final solution quality at the end of the horizon, fit the same framework; the precise shaping is a design choice studied experimentally. The agent thus maximizes the expected cumulative discounted reward over the SA horizon, which aligns the learned proposal with the overall goal of driving the annealing process toward low-cost solutions rather than with any single greedy step.

Node-Centric State Representation

To be applicable to any problem size N and to remain *scale-invariant*, the state is encoded in a *node-centric* fashion:

each node i carries its own local feature vector $\mathbf{x}_i$, and no fixed-size global descriptor is used. The features fall into four groups:

- *Static instance features*: node coordinates, polar coordinates relative to the depot, a depot indicator, and the normalized demand q_i/Q ;
- *Solution topology and local geometry*: the coordinates of the node’s current predecessor and successor in its route, together with local quality signals such as the detour cost incurred by the node at its current position, its distance to the route centroid, and local density statistics of the instance;
- *Capacity status*: the load ratio and remaining slack of the node’s current route, and the node’s own contribution to that load;
- *Search meta-features*: the current (normalized) SA temperature and the fraction of the search budget remaining, which let the policy adapt its behavior to the phase of the annealing schedule.

All features are local to a node and standardized, so the same trained model can be deployed on instances of arbitrary size without retraining. The complete list of features, with their formal definitions, is given in the appendix; their individual contribution is analyzed in the experimental section.

Two-Stage Conditional Policy

The policy must define a distribution over the $\mathcal{O}(N^2)$ node pairs while remaining cheap enough to be evaluated at every SA step. We therefore factorize the joint selection as

$$P(i, j | s) = P(i | s) \cdot P(j | i, s),$$

and implement it with a lightweight two-stage architecture, illustrated in Figure 2. In the first stage, a small network f_θ maps each feature vector to a scalar score $z_i = f_\theta(\mathbf{x}_i)$, and the first node is sampled from the softmax distribution over these scores:

$$i \sim P(\cdot | s) = \text{softmax}(\{z_i : i \in \mathcal{C}\}).$$

In the second stage, the representation of the selected node i is combined with that of every other node j , and a second network g_θ assigns a conditional score $z_{ij} = g_\theta(\mathbf{x}_i, \mathbf{x}_j)$ from which the partner is sampled:

$$j \sim P(\cdot | i, s) = \text{softmax}(\{z_{ij} : j \in \mathcal{C}\}).$$

Both stages are built from small multilayer perceptrons applied identically to every node, which makes the policy permutation-equivariant. The second stage can additionally be conditioned on cheap pairwise descriptors of the candidate move (e.g., the exact cost of the move joining i and j), computed only for the selected i .

This factorization also provides a natural hook for the capacity constraints: once i is selected, the partners j for which $H(s, (i, j))$ would be infeasible are masked out of the second-stage distribution, so every proposed neighbor is feasible at the price of $\mathcal{O}(N)$ feasibility checks per step. Alternative constraint-handling strategies are discussed and compared in the experimental section.

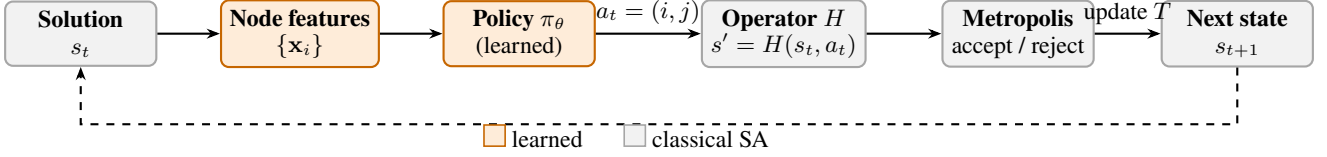


Figure 1: The LG-SA loop. Classical SA components (gray) are kept unchanged: the operator H , the Metropolis acceptance, and the cooling schedule. The only modified block is the move proposal (orange): the current solution is encoded into per-node feature vectors and a learned policy samples the node pair $a_t = (i, j)$, replacing the uniform sampling of classical SA.

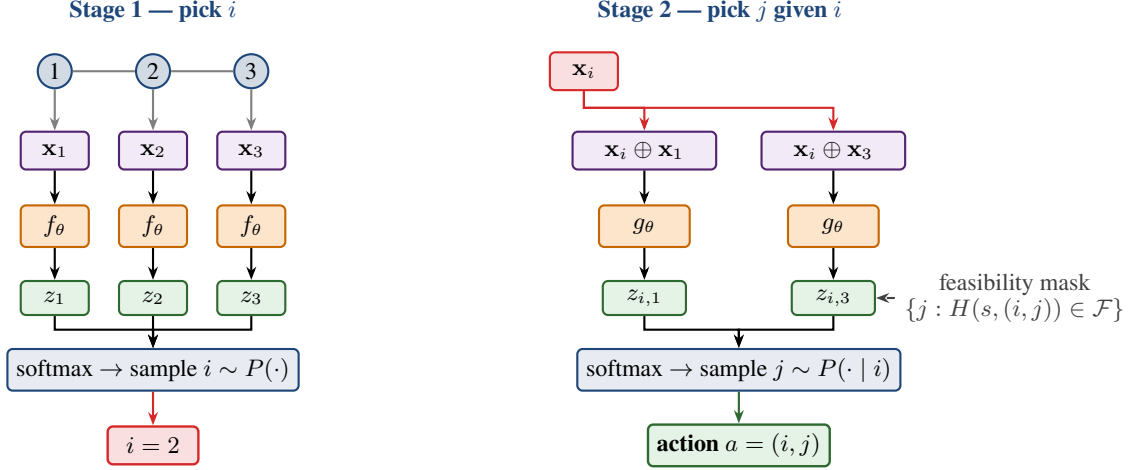


Figure 2: Two-stage conditional sampling of the node pair (i, j) , shown on a toy route of three customers. Stage 1: every node’s feature vector is scored independently by a shared network f_θ and the first node i is sampled from the softmax over the scores. Stage 2: the representation of the selected node i is combined with that of every candidate partner j and scored by a second network g_θ ; infeasible partners are masked out before the softmax, and j is sampled from the conditional distribution. The factorization $P(i, j) = P(i) \cdot P(j | i)$ makes each proposal $\mathcal{O}(N)$.

Since the conditional network is evaluated only for the single selected first node, each proposal costs $\mathcal{O}(N)$ — as opposed to the $\mathcal{O}(N^2)$ cost of scoring all pairs jointly — and the whole step is expressed as batched tensor operations, allowing thousands of instances to be advanced in parallel on one GPU.

Training with PPO

The policy is trained with *Proximal Policy Optimization* (PPO) (Schulman et al. 2017), the standard on-policy actor-critic algorithm. Training alternates between a *collection* phase, in which the current policy runs the full LG-SA loop of Algorithm 1 on a large batch of freshly generated instances in parallel and stores every transition (s_t, a_t, r_t) , and an *update* phase, in which the stored episodes are replayed for several optimization epochs. Advantages are estimated with Generalized Advantage Estimation, and the actor maximizes the usual clipped surrogate objective with an entropy bonus; we refer to Schulman et al. (2017) for the standard formulation and only detail our departures from it.

The critic V_ϕ is a separate permutation-invariant network: it encodes each node feature vector independently, pools the embeddings across nodes into a fixed-size summary, and maps this summary to a scalar value estimate. Keeping the

actor and critic distinct prevents interference between the policy and value representations.

Our implementation departs from standard PPO in three ways, all aimed at stabilizing training on the long and noisy SA horizon:

- *Hybrid clipping–KL objective.* Rather than choosing between the clipped surrogate and the KL-penalized objective, we use both: an adaptive Kullback–Leibler penalty $\beta_{\text{KL}} D_{\text{KL}}(\pi_{\theta_{\text{old}}} || \pi_\theta)$ is added to the clipped loss, and its coefficient is doubled or halved after each update epoch depending on whether the measured divergence leaves a band around a target value. This soft trust region allows many optimization epochs on each collected batch without catastrophic policy drift.
- *KL-coupled learning rate.* The same divergence signal modulates the actor’s learning rate within fixed bounds — gently decreased when the policy moves too fast, increased when it stalls — so the effective step size tracks the trust region automatically.
- *Regularized critic.* The value loss is clipped, mirroring the actor’s trust region, and augmented with a gradient penalty that encourages a smooth (near-Lipschitz) value landscape; we found this useful given that consecutive

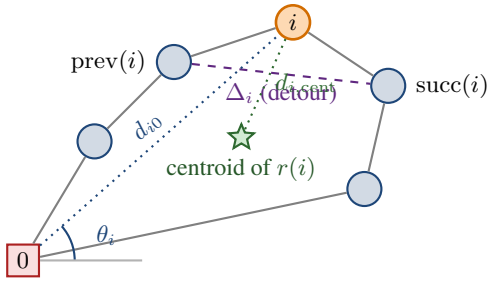


Figure 3: Geometric features attached to a node i (orange) within its current route. The detour Δ_i measures the extra length caused by visiting i between its neighbors; (θ_i, d_{i0}) locate i in polar coordinates around the depot; $d_{i,\text{cent}}$ measures how far i lies from the center of gravity of its own route.

states differ by a single local move whose acceptance is itself stochastic.

Experimental Results

Experimental Setup

Local Search Operators and Feasibility Strategies

Ablation Studies

Comparison with Baselines

Generalization and Scalability

Conclusion

Appendix: State Feature Definitions

This appendix defines every component of the node feature vector $\mathbf{x}_i$ used by the policy and the critic. Throughout, $\text{prev}(i)$ and $\text{succ}(i)$ denote the nodes immediately before and after i in the current visiting order, $r(i)$ the route currently serving i , $Q_{r(i)}$ the total load of that route, and $d(\cdot, \cdot)$ the Euclidean distance. Figure 3 illustrates the geometric quantities. All features are normalized to comparable ranges, and each is computed locally at node i : the encoding contains no fixed-size global descriptor, which is what makes the representation applicable to any instance size.

Static instance features (6). The 2D coordinates $\mathbf{p}_i = (p_i^x, p_i^y) \in [0, 1]^2$; a depot indicator $\mathbf{1}_{\{i=0\}}$; the polar angle θ_i of the node around the depot; the distance to the depot d_{i0} , min-max normalized per instance; and the demand normalized by the vehicle capacity, q_i/Q . These features depend only on the instance, not on the current solution.

Route topology (4). The coordinates $\mathbf{p}_{\text{prev}(i)}$ and $\mathbf{p}_{\text{succ}(i)}$ of the node’s current neighbors in the visiting order. They give the policy direct access to the local shape of the route around each node.

Neighbor distance-ranks (2). For each of $\text{prev}(i)$ and $\text{succ}(i)$, the rank of that neighbor in the list of all nodes sorted by distance from i , normalized to $[0, 1]$ (the nearest node has rank ≈ 0). A well-placed node is linked to low-rank neighbors; a high rank signals a locally poor connection, independently of the absolute scale of the instance.

Group	Features	Dim
Static	$\mathbf{p}_i, \mathbf{1}_{\{i=0\}}, \theta_i, d_{i0}, q_i/Q$	6
Topology	$\mathbf{p}_{\text{prev}(i)}, \mathbf{p}_{\text{succ}(i)}$	4
Neighbor ranks	rank of $\text{prev}(i), \text{succ}(i)$	2
Relative demand	$q_i / \max_j q_j$	1
Spatial density	$\mu_5, \mu_{N/10}, \mu_{N/3}$	3
Local quality	$\Delta_i, d_{i,\text{cent}}$	2
Capacity status	$L_{r(i)}, S_{r(i)}, \eta_i$	3
Meta	temperature, remaining budget	2
Total		23

Table 1: Summary of the node feature vector $\mathbf{x}_i$.

Relative demand (1). The demand normalized by the largest demand in the instance, $q_i / \max_{j \in C} q_j$, complementing q_i/Q with a signal that is invariant to the capacity level.

Spatial density (3). The mean distance from i to its k nearest neighbors, evaluated at three scales: $k = 5$, $k = \lceil N/10 \rceil$, and $k = \lceil N/3 \rceil$. These static statistics distinguish nodes in dense clusters from isolated ones.

Local solution quality (2). The *detour cost*

$$\Delta_i = d(\text{prev}(i), i) + d(i, \text{succ}(i)) - d(\text{prev}(i), \text{succ}(i)),$$

i.e., the extra length the tour pays for visiting i at its current position (equivalently, the saving obtained by removing it); and the distance $d_{i,\text{cent}}$ from i to the centroid of the nodes of its route, which flags spatial outliers assigned to the wrong route.

Capacity status (3). The load ratio of the node’s route, $L_{r(i)} = Q_{r(i)}/Q$; the remaining slack, $S_{r(i)} = (Q - Q_{r(i)})/Q$; and the node’s share of its route load, $\eta_i = q_i/Q_{r(i)}$. These dynamic signals indicate whether a route is nearly saturated and how much a given node contributes to that saturation.

Search meta-features (2). The current SA temperature, normalized to $[0, 1]$ over the schedule range, and the fraction of the search budget remaining. They are identical across nodes of the same state and let the policy modulate its behavior along the annealing schedule (e.g., bolder restructuring at high temperature, fine repairs near the end).

Conditional pair features (stage 2, optional). In addition to the per-node vectors above, the second stage of the policy can receive descriptors of the candidate move itself, computed only for the selected first node u and every candidate partner v : the exact insertion cost $d(v, u) + d(u, \text{succ}(v)) - d(v, \text{succ}(v))$ together with the net cost change of the full move (insertion cost minus the removal saving of u), and the normalized distance-ranks of the edges the move would create. Because they are evaluated for a single u , these descriptors preserve the $\mathcal{O}(N)$ per-step complexity.

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